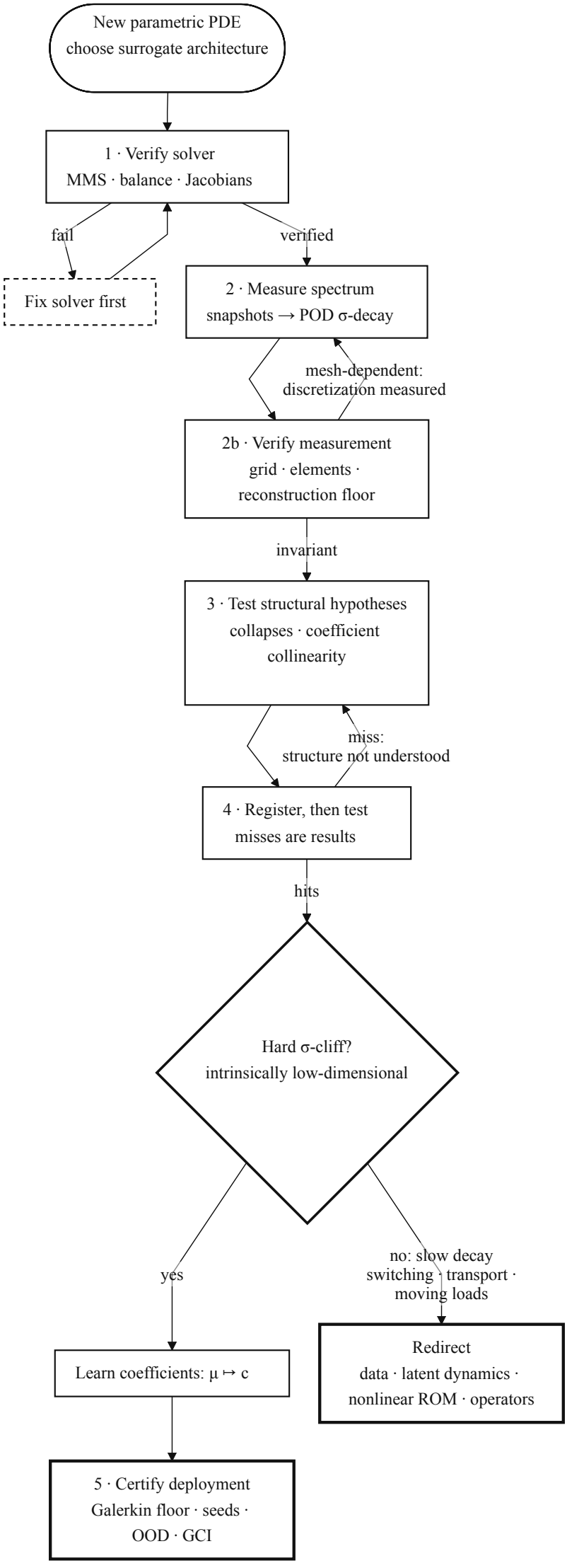
New parametric PDE
choose surrogate architecture
1 · Verify solver
MMS · balance · Jacobians
fail
verified
Fix solver first
2 · Measure spectrum
snapshots → POD σ-decay
mesh-dependent:
discretization measured
2b · Verify measurement
grid · elements ·
reconstruction floor
invariant
3 · Test structural hypotheses
collapses · coefficient
collinearity
miss:
structure not understood
4 · Register, then test
misses are results
hits
Hard σ-cliff?
intrinsically low-dimensional
yes
no: slow decay
switching · transport ·
moving loads
Learn coefficients: μ ↦ c
Redirect
data · latent dynamics ·
nonlinear ROM · operators
5 · Certify deployment
Galerkin floor · seeds ·
OOD · GCI